

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

1. INTRODUCTION

Customer segmentation is a data analysis technique used to divide customers into groups based on similar characteristics. This helps businesses understand customer behavior and improve marketing strategies.

In this project, K-Means Clustering algorithm is used to segment customers based on their annual income and spending score.

2. OBJECTIVES

- To analyze customer data using Python
 - To perform data preprocessing and cleaning
 - To visualize customer behavior patterns
 - To apply K-Means Clustering algorithm
 - To segment customers into meaningful groups
 - To build a Power BI dashboard for visualization
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3. DATASET DESCRIPTION

Dataset Name: Mall Customers Dataset

Source: Kaggle

Features:

- CustomerID: Unique ID for each customer
- Gender: Gender of customer
- Age: Age of customer
- Annual Income (k\$): Annual income in thousand dollars
- Spending Score (1–100): Score assigned based on spending behavior

Total Records: 200 customers

4. TOOLS AND TECHNOLOGIES USED

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Power BI
 - GitHub
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5. DATA PREPROCESSING

The dataset was loaded into Google Colab and the following steps were performed:

- Checked for missing values
 - Verified column names
 - Removed duplicate records (if any)
 - Standardized data for clustering
 - Selected important features for analysis
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6. EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed using visualization techniques:

6.1 Gender Distribution

A bar chart was used to analyze gender distribution among customers.

6.2 Age Distribution

Histogram was used to visualize customer age groups.

6.3 Income Distribution

Distribution of annual income was analyzed.

6.4 Spending Score Distribution

Spending behavior of customers was studied.

6.5 Income vs Spending Score

Scatter plot was used to understand customer segmentation patterns.

7. K-MEANS CLUSTERING

K-Means algorithm was applied to group customers based on:

- Annual Income
- Spending Score

Elbow Method

The Elbow Method was used to determine the optimal number of clusters. The optimal value of K was found to be 5.

Cluster Formation

Customers were grouped into 5 clusters based on similarity.

8. RESULTS AND DISCUSSION

The clustering model identified the following customer groups:

- High income, high spending customers
- High income, low spending customers
- Low income, high spending customers
- Low income, low spending customers
- Average customers

These insights help businesses target customers effectively.

9. POWER BI DASHBOARD

An interactive dashboard was created using Power BI.

Visualizations included:

- Total Customers Card
- Average Age Card
- Average Income Card
- Average Spending Score Card
- Gender Distribution Chart
- Age Distribution Chart
- Income vs Spending Score Scatter Plot
- Cluster Distribution Chart

The dashboard allows interactive filtering based on gender and cluster.

10. CONCLUSION

This project successfully demonstrated customer segmentation using K-Means Clustering. The analysis provided meaningful insights into customer behavior which can help businesses improve marketing strategies and decision-making.

11. REFERENCES

- Kaggle Dataset
 - Scikit-learn Documentation
 - Python Documentation
 - Power BI Documentation
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12. AUTHOR

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Project: Customer Segmentation Using Clustering